Building Our Digital World: Computer Systems & Architecture COMP1860

School of Computer Science, University of Leeds

Activity Sheet 1.3

This worksheet contains a combination of formative activities and activities that contribute towards your portfolio. Portfolio activities are detailed in the section "Portfolio Task". Activities marked by (*) are advanced activities as may take some time to complete.

Expectations:

1. **Timeliness** You should complete all of the activities in the order provided. Activities marked as part of the portfolio should be submitted to Gradescope before the completion date (Friday, 23/10/2026, 14:00).
2. **Presentation:** You should present all of your work clearly and concisely following any additional guidance provided by the module staff in the module handbook.
3. **Integrity:** You are responsible that the evidence you submit as part of your portfolio evidence is entirely your own work. You can find out more about Academic integrity on the Skill@library website. All work you submit for assessment is subject to the academic integrity policy.

Feedback: Feedback on formative activities will be provided via Lab classes and tutorials. Feedback on activities submitted as part of the portfolio will be available on Gradescope.

Support opportunities: Support with the activity sheet is available in the Lab classes and tutorials. Individual support is available via the online booking system.

Expected time for completion: 2-3 hours.

Expected complete date: Friday, 23/10/2026, 14:00

Coursework summary

This worksheet covers material which was introduced in week 3's lectures.

- Representations of non-numeric data.
- Boolean logic.
- Logic circuits.

Learning outcomes

On completion of this activity sheet you will:

1. Explain the bit-level representations of non-numeric data.
2. Construct a logic circuit using logic gates.

Portfolio Task

The portfolio task is a Gradecope online quiz. You will find a link to it in the COMP1860 Minerva module.

Formative Questions

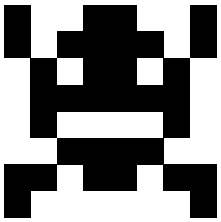
Non-numeric Data Representations

Character Representations

1. Convert the word **University** to binary using the ASCII encoding. **Hint:** Convert each letter in the word to its ASCII binary representation.
2. (*) The Unicode code point for a "beaming face with smiling eyes" emoji is U+1F601.
   (a) Calculate the binary encoding in UTF-8.
   (b) Give the hexadecimal encoding by converting your binary encoding.

Image Representations

1. Create a NetPBM file to encode the following image (a space invader):



Boolean Logic & Logic Circuits

1. What is the ANSI/IEEE logic circuit symbol used to denote the following gates?
   (a) AND
   (b) NOT
   (c) OR
   (d) NAND
   (e) XOR
   (f) NOR
2. For each of the following expressions construct:
   - a truth table
   - a logic circuit diagram (use ANSI/IEEE symbols)

   (a) $x \wedge s$
   (b) $y \wedge \neg s$
   (c) $(x \wedge s) \vee (y \wedge \neg s)$
   (d) $\neg(x \wedge y)$
   (e) $\neg x \vee y$

(f) $(x \vee y) \wedge \neg(x \wedge y)$

3. (*) For each of the the expressions in the previous question construct:

   - an equivalent expression in conjunctive normal form (CNF)
   - an equivalent expression in disjunctive normal form (DNF)